MATH 141
Calculus With Analytic Geometry II
Fall 2025

1 Course information

1.1 Catalogue Description

MATH 141 (GQ) Calculus With Analytic Geometry II (4 credits). MATH 141 is the second course in a two- or three-course calculus sequence for students in science, engineering and related fields. Calculus is an important building block in the education of any professional who uses quantitative analysis. This course further introduces and develops the mathematical skills required for analyzing growth and change and creating mathematical models that replicate reallife phenomena. The goals of our calculus courses include to develop the students' knowledge of calculus techniques and to use the calculus environment to develop critical thinking and problem solving skills. This course covers the following topics: logarithms, exponentials, and inverse trigonometric functions; applications of the definite integral and techniques of integration; sequences and series; power series and Taylor polynomials; parametric equations and polar functions. Students may take only one course for credit from MATH 141, 141B, and 141H.

1.2 Enforced Prerequisite

MATH 140 or MATH 140B or MATH 140E or MATH 140G or MATH 140H.

1.3 Required Materials

1. Textbook: Calculus: Early Transcendentals by Jon Rogawski, Colin Adams, and Robert Franzosat. 4th ed. 2019. Macmillan.

2. Access to the online homework platform.

1.3.1 Options for Purchasing the Required Materials

The required materials are available in these different forms (you need to purchase only one):

1. Electronic Package: ISBN 9781319371883. This option provides access to the ebook version of the course materials through a single Achieve account.

2. Print Package: ISBN 9781319371869 This option includes a hardback edition of the textbook and access to the ebook. **Note:** The print option includes an activation code for the ebook as well, so anyone who purchases the print option will also have access to the ebook version through their Achieve account.

3. Loose-Leaf Package: ISBN 9781319371845. This option includes a printed loose-leaf edition of the textbook and access to the ebook. Note: As with the Print option the loose-leaf option includes an activation code for the ebook as well, so anyone who purchases the loose-leaf option will also have access to the ebook version through their Achieve account. Where to Purchase the Required Materials: PSU Bookstore or directly from the Publisher.

Important Notes:

- Please do not purchase or rent any edition of the textbook through any source not listed above. Currently, only the publisher and the University Bookstore offers the edition of the textbook used in this class. Please check with your instructor before buying or renting through another source.
- Regardless of how you purchase access, you **must access the Achieve link via Canvas, and you must use your PSU email to set up your Macmillan Account.**

2 Mode of Instruction

Classes will be held in-person per University Policy. The in-person lectures are not to be recorded, and if a student misses a lecture, they should communicate with their instructor and ask classmates for notes. The student will also need to read the textbook and consult the instructor for help and support.

3 Student Absence

Students who are unable to attend class or complete graded assignments for more than one week at a time will be required to present documentation to verify their significant, prolonged illness. From UHSPolicies & Patient Resources: "For routine illness-related absences, students should correspond directly with the faculty as soon as possible regarding their situation, ideally, before they miss a class, exam, or other evaluative activity. University Health Services may provide verification of illness forms for significant prolonged illnesses or injuries lasting at least a week resulting in absence from classes. When appropriate, students may request the verification during their clinician visit or send a secure message to their clinician or the Advice Nurse through myUHS at **https://studentaffairs.psu.edu/health/myuhs**. If a student wants verification of illness from University Health Services and has received care from an outside provider for a significant, prolonged illness, they must provide appropriate documentation to the University Health Services director, 502A StudentHealth Center, 814-865-6555."

4 Course Goals and Objectives

4.1 Course Description

This course satisfies the following General Education Qualification.

1. Effective Communication
2. Key Literacies
3. Critical and Analytic Thinking
4. Integrative Thinking

4.2 Course Goals

At the end of the semester, students should be able to:

1. Apply techniques of integration in various settings, including solving basic differential equations.
2. Generate infinite series representation of functions and use them to evaluate limits, derivatives, and integrals.
3. Construct complex power series representations and the exponential representation of complex numbers.

4.3 Learning Objectives

The detailed objectives that correspond to the goals above are as follows.

1. Apply techniques of integration in various settings, including solving basic differential equations.
   (a) Apply methods of integration, such as substitution (u-sub and trigonometric sub), integration by parts, and partial fraction decomposition to evaluate antiderivatives of functions (algebraic, trigonometric, inverse trigonometric, hyperbolic, and transcendental functions).
   (b) Apply the Fundamental Theorem of Calculus to evaluate definite integrals.
   (c) Develop knowledge of basic differential equations, stable points, and graphs of solutions.
   (d) Apply the basic knowledge of differential equations to write simple differential equation models and investigate the solutions qualitatively.
   (e) Apply the knowledge of integration techniques to solve and interpret solutions to separable differential equations.
   (f) Solve and interpret solutions to differential equation models such as Exponential Growth and Exponential Decay, Logistic Growth, Newton's Law of Cooling, Torriccelli's Law, and so on.
   (g) Evaluate limits of functions through L'Hôpital's rule to compare relative rates of growth or asymptotic behavior of functions.
   (h) Use limits to asymptotically analyze the behavior of solutions to differential equations.
   (i) Use algebraic techniques and L'Hôpital's rule to compute limits of improper integrals.
   (j) Solve improper integrals and interpret the meaning of convergence and divergence while emphasizing the notion of accumulation.
   (k) Apply the concept of improper integration to learn about probabilities, Laplace Transforms, and other applications of improper integrals.
   (l) Fluently recognize appropriate integration techniques, apply and compute integrals, and interpret the meaning of integration in a multitude of contexts.

2. Generate infinite series representation of functions and use them to evaluate limits, derivatives, and integrals.
   (a) Understand the different types of a sequence of terms, evaluate the limits of sequences, and explain what it means for a sequence to convergence or diverge.
   (b) Connect the concept of sequences to that of infinite series.
   (c) Analyze when an infinite series converges or diverges by applying an appropriate test(s).
   (d) Develop fluency in choosing appropriate test(s), when analyzing convergence and divergence of infinite series with positive terms and otherwise.
   (e) Express functions as infinite series using Geometric series or Powers series in general, with an emphasis on the interval of convergence and the radius of convergence.
   (f) Use Taylor polynomials to approximate functions and determine the error associated with the estimation.
   (g) WriteTaylor series (including McLaurin series) representations of functions and address the notion of the remainder between a function and its representation.
   (h) Apply Power series and Taylor series representations of functions to find limits, derivatives, integrals, and solutions to differential equations.

(i) Apply Power series representations to obtain complex power series and to obtain polar coordinate

3. Construct complex power series representations and the exponential representation of complex numbers.
   (a) Discuss polar coordinates and how to represent Cartesian coordinates.
   (b) Develop fluency in switching between polar coordinates and Cartesian coordinates.
   (c) Apply power series representations to obtain complex power series.
   (d) Use power series representation to obtain the exponential form of complex numbers.

5 Calculators

A calculator or any other technological tool such as Desmos, Geogebra, WolframAlpha, and so on are an excellent tool for crunching numbers and generating plots especially when working real-world examples as well as a useful study tool. **Note, however, that calculators are not necessary to understand and apply the concepts of the course to examples with simple coefficients. Calculators will not be allowed during quizzes or exams (all arithmetic computations will be simple enough to do by hand)**.

6 Tutors and Math Center

Free mathematics tutoring is available at Penn State Learning located in 007 Sparks Building. For more information, go to the https://pennstatelearning.psu.edu/tutoring/mathematics.

7 Assessments

7.1 Summative Assessments (450 points)

During the semester, this course will have **four total examinations**. Three of these examinations will occur during the semester (each examination will be 75-minutes long and worth 100 points each) and a **cumulative final exam** will occur during the week of finals (final exam is 110-minutes long and worth 150 points).

Exam rooms will be announced later. **No books and notes are allowed on examinations. You must bring your University ID card to all exams**. The room information for the exams will be announced on CANVAS. It is not permissible to take the exam in a different instructor's assigned room.

These exams will be graded on a partial credit scale for correctness and the explanations of the solutions. An answer with no work will not receive credit. **Using notes or a calculator are not allowed, and seeking help from other students during an exam are grounds for an academic integrity violation and will result in consequences in accordance with** G-9: Academic Integrity

Examination I	09/25/2025	6:15 P.M. - 7:30 P.M.
Examination II	10/23/2025	6:15 P.M. - 7:30 P.M.
Examination III	11/17/2025	6:15 P.M. - 7:30 P.M.
Final Examination	During the Week of Finals	During the Week of Finals

Authorized Early Exam& Makeup Exam Policies In addition to the regularly scheduled midterm examinations, the math department provides two options for students who are unable to take the midterm at the scheduled time: a authorized early exam and a makeup exam.

- A authorized early exam for each of the midterms is offered on the same night as the regularly scheduled exam at a different time. **Students who attend the authorized early exam will not be permitted to leave the exam room until 6:10 pm**.
- A makeup exam is scheduled on an evening different from that of the regularly scheduled exam.

In order to qualify for either the conflict or makeup exam, you must have a valid conflict or reason to take the makeup. If you think you will not be able to take the regularly scheduled exam and would like to take either the conflict or makeup exam, you must

1. Contact your instructor to determine if you are eligible to take a conflict or makeup exam,
2. Provide all necessary documentation where relevant or necessary,
3. Receive confirmation from your instructor that you are eligible for a conflict or makeup exam, and
4. Sign up for the conflict or makeup exam using the online signup form provided in CANVAS no later than the specified deadline (also listed in CANVAS).

Note: Students who miss a regular, conflict, or makeup exam without a valid reason and the approval of their instructor will receive a zero.

Who May Take The Authorized Early Exam? If you have a valid conflict with the regular examination time, such as a class or another official University activity, you may sign up for the authorized early exam. If a student has not signed up for the authorized early exam, they will not be permitted to take the exam.

Instructions On Authorized Early Exam Night. The student is responsible for knowing the room and time of the authorized early examination. Each student must bring his or her University ID to the authorized early examination. The ID will be checked by the proctor. Although the authorized early examination will end at 6:05 pm, no student will be permitted to leave the examination room before 6:10 pm. **Any student who leaves before 6:10 pm will receive a grade of zero on the examination and will not be allowed to retake it**.

Who May Take the Makeup Exam? Students who have a valid University reason (University-approved curricular and extra-curricular activities) such as a class conflict or illness, during both the conflict and regular examination times are permitted to schedule a makeup examination with no penalty. Students must be prepared to verify the reason for taking the makeup and may be asked to provide documentation. **Students who do not have a valid reason for missing the examination are permitted to schedule the makeup, but at a 25% penalty**. Students who have taken either

Makeup Examination I	10/01/2025	6:15 P.M. - 7:30 P.M.
Makeup Examination II	10/29/2025	6:15 P.M. - 7:30 P.M.
Makeup Examination III	12/02/2025	6:15 P.M. - 7:30 P.M.

the regularly scheduled examination or the authorized early examination are not permitted to take the makeup examination. The makeup examinations are given on the evenings and times listed below:

How and When to Sign up for the Makeup Exam A student who is ill on exam night must contact his or her instructor within 24 hours of the exam. Students must sign up for the makeup exam at the provided signup link (in CANVAS), as soon as possible following the regular exam date. The online signup forms for both the authorized early exam and the makeup exam are available in CANVAS. The student is responsible for knowing the room and time of the makeup examination. If a student has not signed up with his or her instructor, the student will not be allowed to take the makeup exam.

Instructions On Makeup Exam Night The student is responsible for knowing the room and time of the makeup examination. Each student must bring his or her PSU ID to the makeup examination. The ID will be checked by the proctor.

What If A Student Misses Both The Regularly Scheduled Exam and The Makeup Exam? If a student misses both the regularly scheduled examination and the scheduled makeup due to a valid, verifiable reason, it may be possible to take a makeup examination by appointment. All such makeup examinations must be scheduled through the classroom instructor and must be completed no later than one week after the scheduled makeup examination.

What If A Student Misses All Exam Possibilities? If a student misses the regularly scheduled examination, the authorized early exam, the scheduled makeup exam, and the makeup to the makeup option, then the student will receive a zero on that exam. Unless a student has a valid University-approved absence and can provide all necessary documentation to support their absence.

Final Examination: The final examination will be given during the week of **Monday December 15th - Friday December 19th, 2025**. **The final examination may be scheduled on any day during the final examination period. Do not plan to leave University Park until after Friday December 19th, 2025**.

Notification of conflicts is given on the students final exam schedule. There are two types of authorized early examinations: direct and overload. Direct conflicts are two examinations scheduled at the same time. Overload examinations are three or more examinations scheduled within a fifteen-hour period, from the beginning of the first examination to the beginning of the third examination. Students may elect to take three or more examinations on the same day if they wish or request a conflict final examination.

A student must take action to request a authorized early exam through LionPATH between **September 29th through October 26th**. **Conflict final examinations cannot be scheduled through the Mathematics department, and there will be no sign-up sheet in class for the final authorized early examination**. Students who miss or cannot take the final examination due to a valid and documented reason, such as illness, may be allowed to take a makeup final examination at the beginning of the next semester. **Early flights home, bus tickets to leave town, and family vacations are NOT valid excuses to miss or reschedule a final exam. If the student does not have a valid reason, and if approved by the course coordinator, may be allowed to take a conflict final exam, but a**

30% penalty will be imposed. All such makeup examinations must be arranged through the instructor with the approval of the course coordinator, and students in such a situation should contact their instructor within 24 hours of the scheduled final examination. Students who have taken the original final examination are not permitted to take a makeup examination.

7.2 Formative Assessments (300 Points)

1. **Written homework (100 Points)** sets will consist of a set of problems, which will be posted on CANVAS. Fully worked-out solutions to the homework problems must be uploaded to Canvas or GradeScope. Solutions must be clear, legible, have substance, and all answers must be circled. **A solution with no work, will receive no credit. Also, all application questions must include an interpretation, which might be 1 to 2 sentences long, and appropriate units**. Due dates and times will be posted on CANVAS.

2. **Online Homework (100 points)** needs to be accessed and completed online through **Achieve**. Each problem on the online homework will have 10 attempts to complete, and everyone is allowed to submit the online homework up to 2 days late, but with a 20% penalty.

3. **In-class Quizzes (100 Points)** will be scheduled weekly and will occur in-class. Quizzes will cover the same topics as the homework (online and written homework) of that week.

7.3 Evening LA Sessions

Each week, the Learning Assistants (LAs), who are part of the instructional team, will hold multiple sessions. Attending these sessions is not required, but strongly recommended. These sessions are a continuation of what we will be discussing in class. By attending these sessions, you (the student) are strengthening your skills and understanding of the content. More information about the general structure, time, and location will be communicated later via CANVAS. To incentivize students to attend, the following will be applied:

- If a student attends at least 5 evening LA sessions out of the 11 possible evening LA sessions during the Exam 1 period, the student will be able to replace their lowest in-class quiz score during that exam time period with a 100% (10 out of 10).

- If a student attends at least 5 evening LA sessions out of the 11 possible evening LA sessions during the Exam 2 period, the student will be able to replace their lowest in-class quiz score during that exam time period with a 100% (10 out of 10).

- If a student attends at least 4 evening LA sessions out of the 9 possible evening LA sessions during the Exam 3 period, the student will be able to replace their lowest in-class quiz score during that exam time period with a 100% (10 out of 10).

- If a student attends at least 3 evening LA sessions out of the 8 possible evening LA sessions during the Final Exam period, the student will be able to replace their lowest in-class quiz score during that exam time period with a 100% (10 out of 10).

8 Course Policies, Valid Excuses, and Grading Policy

8.1 Course Policies

a. One lowest written homework, one lowest online homework, and one lowest in-class quiz will be dropped. NO exam score will be dropped.

b. There are no extra credit assignments available for any student to earn extra points in order for them to improve their course overall grade.

c. Makeup quizzes and homework extensions (written and/or online) are only given if a student has a valid excuse and informs their course instructor by email of their absence and inability to participate the day prior to the quiz or the day prior to when the homework (written and/or online) is due. **However, the instructor has the right to decide on makeups as they see fit**. The student and the instructor must make the appropriate arrangements for an extension on the homework and to schedule a makeup quiz.

d. If a student often requests makeups and extensions, **more than two times**, the course instructor has the right to deny all such requests unless the student has a University-approved reason and can provide proper documentation when necessary.

e. If a student is scheduled for a makeup quiz and does not show, the instructor has the right to give the student a zero unless the student has a University-approved reason and can provide proper documentation when necessary.

f. All makeup quizzes and homework (online and/or written) extensions must be completed promptly. Preferably, students should complete their makeup quiz and assignments within 24 - 72 hours from the regular quiz and homework due dates. In case of illness or prolonged circumstances, students will receive a makeup quiz when they can but may be required to present University-approved documentation. The student and the instructor must make all appropriate arrangements to schedule all makeup requests.

g. Questions regarding the grading of exams or missing exam grades should be directed to your section instructor as soon as possible. **Students and instructors will have until the next exam for the course takes place to review a physical copy of the exam or report a missing grade**. Students will have one week from the date of the final exam to contact their instructor regarding questions about the final exam or a missing final exam grade. After each midterm examination, the student will have until the next exam to bring any issues about the grades or grading up to their course instructor for review. This means that

   - issues about exam 1 grade or grading should be brought up to the course instructor's attention before exam 2,
   - issues about exam 2 grade or grading should be brought up to the course instructor's attention before exam 3,
   - and issues about exam 3 grade or grading should be brought up to the course instructor's attention before the last day of classes.

   All paper copies of the exams will be shredded right before the next exam. Therefore, students must fully review their examination grades and grading after the exams are released on GradeScope. Any late requests will not be considered.

8.2 Exam Policies and Valid Excuses

During the course, many possible situations arise that result in your inability to attend class, attend exams, or perform at a minimally acceptable level during an examination. Illness or injury, family emergencies, certain University-approved curricular and extra-curricular activities, and religious holidays can be legitimate reasons to miss class or to be excused from a scheduled examination.

a. With regard to family emergencies, you must provide verifiable documentation of the emergency. Given the vast array of family emergencies, the instructor will provide precise guidance as to what constitutes adequate documentation. Unless the emergency is critical, you should notify the instructor in advance of your absence from the scheduled course event. In cases of critical emergencies, you must notify the instructor within one week of your absence.

b. For University-approved curricular and extra-curricular activities, verifiable documentation is also required. The student should obtain from the unit or department sponsoring the activity a letter (or class absence form) indicating the anticipated absence(s). The letter must be presented to the instructor at least one week prior to the first absence.

c. In the case of religious holidays, the student should notify the instructor by the third week of the course of any potential conflicts.

d. Students who have a valid University reason, such as a class conflict or illness, during both the conflict and regular examination times are permitted to schedule a makeup examination with no penalty. Students must be prepared to verify the reason for taking the makeup and may be asked to provide documentation. **Students who do not have a valid reason for missing the examination are permitted to schedule the makeup, but at a 25% penalty**. Students who have taken either the regularly scheduled examination or the authorized early examination are not permitted to take the makeup examination.

e. A student who is ill on exam night must contact their course instructor within 24 hours of the exam. Students must sign up for the makeup exam at the provided signup link (in CANVAS), as soon as possible following the regular exam date. The online signup forms for both the authorized early exam and the makeup exam are available in CANVAS.

f. If a student misses the regularly scheduled examination and the regularly scheduled makeup due to a valid, verifiable reason, it **may be possible to have one opportunity to take a makeup examination by appointment**. All such makeup examinations must be scheduled through the classroom instructor and must be completed no later than one week after the scheduled makeup examination.

g. If a student misses the regularly scheduled examination, the authorized early exam, the scheduled makeup exam, and the makeup to the makeup option, then the student will receive a zero on that exam. Unless a student has a valid University-approved absence and can provide all necessary documentation to support their absence.

h. An important note for final exams: **Early flights home, bus tickets to leave town, and family vacations are NOT valid excuses to miss or reschedule a final exam**. Students should make plans to leave campus AFTER all their scheduled exams are completed. It is best not to book flights that leave during finals week. Instructors must give the final exams according to the University schedule and cannot give makeups or reschedules for non-valid or non-approved excuses. The final examination will be given during the week of **Monday December 15th - Friday December 19th and the final exam may be scheduled on any day during the final exam period. Do not plan to leave University Park until after Friday December 19th, 2025**.

8.3 Grading Policy and Final Course Grades

Formative Assessment	300 points
Written Homework	100 points
Online Homework	100 points
In-class Quizzes	100 points
Summative Assessments	450 points
Examination I	100 points
Examination II	100 points
Examination III	100 points
Final Examination	150 points
Total Possible Points	750 points

Final course grades will be assigned as follows

A	92.5% to 100%
A-	89.5% to $<$ 92.5%
B+	86.5% to $<$ 89.5%
B	82.5% to $<$ 86.5%
B-	79.5% to $<$ 82.5%
C+	76.5% to $<$ 79.5%
C	69.5% to $<$ 76.5%
D	59.5% to $<$ 69.5%
F	0% to $<$ 59.5%

9 General University Policies

9.1 LATE-DROP

Students may add/drop a course without academic penalty within the first six calendar days of the semester. A student may late drop a course within the first twelve weeks of the semester. After the first six days and before deciding to late drop this course, each student should consult with their academic advisor. The late drop deadline for Fall 2025 is **Friday, November 14th at 11:59 p.m. (ET)**.

9.2 Deferred Grades

Students who are currently passing a course but are unable to complete the course because of illness or emergency may be granted a deferred grade which will allow the student to have up to ten weeks from the final grade reporting deadline to complete the course. Note that deferred grades are limited to those students who can verify and document a valid reason for not being able to take the final examination.

Approval needs to be granted prior to the beginning of the final exam period of the semester in which the course is taken. For more information see https://www.registrar.psu.edu/grades/deferred.cfm

9.3 Academic Integrity

Academic integrity is the pursuit of scholarly activity in an open, honest and responsible manner. Academic integrity is a basic guiding principle for all academic activity at The Pennsylvania State University, and all members of the University community are expected to act in accordance with this principle. Consistent with this expectation, the University's Code of Conduct states that all students should act with personal integrity, respect other students' dignity, rights and property, and help create and maintain an environment in which all can succeed through the fruits of their efforts. Academic integrity includes a commitment not to engage in or tolerate acts of falsification, misrepresentation or deception. Such acts of dishonesty violate the fundamental ethical principles of the University community and compromise the worth of work completed by others.

According to Penn State policy G-9: Academic Integrity, an academic integrity violation is an intentional, unintentional, or attempted violation of course or assessment policies to gain an academic advantage or to advantage or disadvantage another student academically. Unless your instructor tells you otherwise, you must complete all course work entirely on your own, using only sources that have been permitted by your instructor, and you may not assist other students with papers, quizzes, exams, or other assessments. If your instructor allows you to use ideas, images, or word phrases created by another person (e.g., from Course Hero or Chegg) or by generative technology, such as ChatGPT, you must identify their source. You may not submit false or fabricated information, use the same academic work for credit in multiple courses, or share instructional content. Students with questions about academic integrity should ask their instructor before submitting work. Students facing allegations of academic misconduct may not drop/withdraw from the affected course unless they are cleared of wrongdoing (see G-9: Academic Integrity). Attempted drops will be prevented or reversed, and students will be expected to complete course work and meet course deadlines. Students who are found responsible for academic integrity violations face academic outcomes, which can be severe, and put themselves at jeopardy for other outcomes which may include ineligibility for Deans List, pass/fail elections, and grade forgiveness. Students may also face consequences from their home/major program and/or The Schreyer Honors College.

9.4 Disability Services

Penn State welcomes students with disabilities into the University's educational programs. Every Penn State campus has an office for students with disabilities. Student Disability Resources (SDR) website provides contact information for every Penn State campus http://equity.psu.edu/sdr/disability-coordinator. For further information, please visit Student Disability Resources website: http://equity.psu.edu/sdr/. In order to receive consideration for reasonable accommodations, you must contact the appropriate disability services office at the campus where you are officially enrolled, participate in an intake interview, and provide documentation: See documentation guidelines http://equity.psu.edu/sdr/guidelines. If the documentation supports your request for reasonable accommodations, your campus disability services office will provide you with an accommodation letter. Please share this letter with your instructors and discuss the accommodations with them as early as possible. You must follow this process for every semester that you request accommodations.

9.5 Educational Equity and Report Bias

Consistent with University Policy AD29, students who believe they have experienced or observed a hate crime, an act of intolerance, discrimination, or harassment that occurs at Penn State are urged to report these incidents as outlined on the Universitys Report Bias webpage http://equity.psu.edu/reportbias/

9.6 Counseling and Psychological Services

Many students at Penn State face personal challenges or have psychological needs that may interfere with their academic progress, social development, or emotional wellbeing. The University offers a variety of confidential services to help you through difficult times, including individual and group counseling, crisis intervention, consultations, online chats, and mental health screenings. These services are provided by staff who welcome all students and embrace a philosophy respectful of clients cultural and religious backgrounds, and sensitive to differences in race, ability, gender identity and sexual orientation.

- Counseling and Psychological Services at University Park (CAPS) http://studentaffairs.psu.edu/counseling/
- Counseling and Psychological Services at Commonwealth Campuses https://senate.psu.edu/faculty/counseling-services-at-commonwealth-campuses/
- Penn State Crisis Line (24 hours/7 days/week): 877-229-6400 Crisis Text Line (24 hours/7 days/week): Text LIONS to 741741

Questions, Concerns, or Comments: If you have questions or concerns about the course, **please consult your instructor first**. If further guidance is needed, you may contact the course coordinator, whose contact information is given below.

Coordinator Contact Information
Amine Benkiran
Associate Teaching Professor
Calculus Coordinator (Math 041, Math 140, and Math 141)
104-H McAllister Building
University Park, PA 16802
Email: azb165@psu.edu

To better assist you, the student, in your email to the coordinator you must include all of the following; otherwise, your email will not be addressed:

- Full Name
- Class
- Section number

When emailing the course coordinator, expect an answer within 24 to 48 hours. Also, all communications must be through a PSU email address such as abc123@psu.edu

Week	Day	Date	Sections	Sections	Important Dates	Achieve and Written Homework
1	Monday	25-Aug	Introduction			
	Tuesday	26-Aug				
	Wednesday	27-Aug	Review of Basic Integrals and and substitution			
	Thursday	28-Aug	Review of Basic Integrals and and substitution			
	Friday	29-Aug	**Intro to D.Es**: verify solutions, equilibria& stability, sketch of general solutions, intro to separable D.Es.	9.1, 9.2, and 9.4		
	Saturday	30-Aug			Regular Drop Deadline	
	Sunday	31-Aug			Regular Add Deadline, Late Drop Begins	
2	Monday	1-Sep	**Labor Day - No Class**		Late Registration Begins	
	Tuesday	2-Sep				
	Wednesday	3-Sep	**Intro to D.Es**: verify solutions, equilibria & stability, sketch of general solutions, intro to separable D.Es.	9.1, 9.2, and 9.4		
	Thursday	4-Sep	D.Es and substitution (u-sub and trig. Integrals)	5.7, 7.2, and 9.2		
	Friday	5-Sep	D.Es and substitution (u-sub and trig. Integrals)	5.7, 7.2, and 9.2		
	Saturday	6-Sep				Achieve: Integration Review and 9.1
	Sunday	7-Sep				Written 1: Integration Review and 9.1
3	Monday	8-Sep	D.Es and substitution (u-sub and trig. Integrals)	5.7, 7.2, and 9.2		Quiz 1: Integration Review and 9.1
	Tuesday	9-Sep				
	Wednesday	10-Sep	D.Es and Integration by Parts	7.1		
	Thursday	11-Sep	D.Es and Integration by Parts	7.1		
	Friday	12-Sep	Trigonometric Substitution	7.3		
	Saturday	13-Sep				Achieve: 5.7, 7.2, and 9.2
	Sunday	14-Sep				Written 2: 5.7, 7.2, and 9.2
4	Monday	15-Sep	Trigonometric Substitution	7.3		Quiz 2: 5.7, 7.2, and 9.2
	Tuesday	16-Sep				
	Wednesday	17-Sep	Trigonometric Substitution	7.3		
	Thursday	18-Sep	Applications of D.Es.	9.2		
	Friday	19-Sep	Applications of D.Es.	9.2		
	Saturday	20-Sep				Achieve: 7.1, 7.3, and applications
	Sunday	21-Sep				Written 3: 7.1, 7.3, and applications
5	Monday	22-Sep	Exam 1 Review			Quiz 3: 7.1, 7.3, and applications
	Tuesday	23-Sep				
	Wednesday	24-Sep	Exam 1 Review			
	Thursday	25-Sep	O.D.Es and Partial Fractions Decomposition	9.4 and 7.5	Exam 1: Sections 5.7, 7.1, 7.2, 7.3, 9.1, and 9.2.	
	Friday	26-Sep	O.D.Es and Partial Fractions Decomposition	9.4 and 7.5		
	Saturday	27-Sep				
	Sunday	28-Sep				
6	Monday	29-Sep	O.D.Es and Partial Fractions Decomposition	9.4 and 7.5	Final Exam Conflict Filing Period Starts on LionPath	Achieve: 9.4 and 7.5
	Tuesday	30-Sep				Written 4: 9.4 and 7.5
	Wednesday	1-Oct	L'Hopital's Rule and Relative Rates of Growth	4.5		Quiz 4: 9.4 and 7.5
	Thursday	2-Oct	L'Hopital's Rule and Relative Rates of Growth	4.5		

Week	Day	Date	Sections	Sections	Important Dates	Achieve and Written Homework
	Friday	**3-Oct**	L'Hopital's Rule and Relative Rates of Growth	4.5		
	Saturday	**4-Oct**				
	Sunday	**5-Oct**				
7	**Monday**	**6-Oct**	Improper Integrals and Applications	7.7 and 8.1		Achieve: 4.5
	Tuesday	**7-Oct**				Written 5: 4.5
	Wednesday	**8-Oct**	Improper Integrals and Applications	7.7 and 8.1		Quiz 5: 4.5
	Thursday	**9-Oct**	Improper Integrals and Applications	7.7 and 8.1		
	Friday	**10-Oct**	Taylor Polynomials and Function Approximation	10.7		
	Saturday	**11-Oct**				
	Sunday	**12-Oct**				
8	**Monday**	**13-Oct**	Taylor Polynomials and Function Approximation	10.7		Achieve: 7.7 and 8.1
	Tuesday	**14-Oct**				Written 6: 7.7 and 8.1
	Wednesday	**15-Oct**	Taylor Polynomials and Function Approximation	10.7		Quiz 6: 7.7 and 8.1
	Thursday	**16-Oct**	Sequences	10.1		
	Friday	**17-Oct**	Sequences	10.1		
	Saturday	**18-Oct**				
	Sunday	**19-Oct**				
9	**Monday**	**20-Oct**	Sequences	10.1		Achieve: 10.7 and 10.1
	Tuesday	**21-Oct**				Written 7: 10.7 and 10.1
	Wednesday	**22-Oct**	Exam 2 Review			Quiz 7: 10.7 and 10.1
	Thursday	**23-Oct**	Exam 2 Review		Exam 2: Sections 9.4, 7.5, 4.5, 7.7, 8.1, 10.7, and 10.1.	
	Friday	**24-Oct**	Series	10.2		
	Saturday	**25-Oct**				
	Sunday	**26-Oct**			Final Exam Conflict Filing Period Ends on LionPath	
10	**Monday**	**27-Oct**	Series	10.2		
	Tuesday	**28-Oct**				
	Wednesday	**29-Oct**	Series	10.2		Achieve: 10.2
	Thursday	**30-Oct**	Convergence of Series with Positive Terms	10.3		Written 8: 10.2
	Friday	**31-Oct**	Convergence of Series with Positive Terms	10.3		Quiz 8: 10.2
	Saturday	**1-Nov**				
	Sunday	**2-Nov**				
11	**Monday**	**3-Nov**	Convergence of Series with Positive Terms	10.3		Achieve: 10.3
	Tuesday	**4-Nov**				Written 9: 10.3
	Wednesday	**5-Nov**	Absolute and Conditional Convergence	10.4		Quiz 9: 10.3
	Thursday	**6-Nov**	Absolute and Conditional Convergence	10.4		
	Friday	**7-Nov**	Absolute and Conditional Convergence	10.4		
	Saturday	**8-Nov**				
	Sunday	**9-Nov**				
	Monday	**10-Nov**	Absolute and Conditional Convergence	10.4		Achieve: 10.4
	Tuesday	**11-Nov**				Written 10: 10.4
	Wednesday	**12-Nov**	Ratio and Root Test	10.5		Quiz 10: 10.4

Week	Day	Date	Sections	Sections	Important Dates	Achieve and Written Homework
12	Thursday	13-Nov	Ratio and Root Test	10.5		
	Friday	14-Nov	Exam 3 Review		Late Drop - Deadline at 11:59 P.M.	
	Saturday	15-Nov				Achieve 10.5
	Sunday	16-Nov				Written 11: 10.5
13	Monday	17-Nov	Exam 3 Review		Exam 3: Sections 10.2, 10.3, 10.4, and 10.5.	
	Tuesday	18-Nov				
	Wednesday	19-Nov	Power Series	10.6		
	Thursday	20-Nov	Power Series	10.6		
	Friday	21-Nov	Power Series	10.6		
	Saturday	22-Nov				
	Sunday	23-Nov				
	Monday	24-Nov	**Thanksgiving Break**			
	Tuesday	25-Nov				
	Wednesday	26-Nov				
	Thursday	27-Nov				
	Friday	28-Nov				
	Saturday	29-Nov				
	Sunday	30-Nov				
14	Monday	1-Dec	Taylor Series	10.8		Achieve 10.6
	Tuesday	2-Dec				Written 12: 10.6
	Wednesday	3-Dec	Taylor Series	10.8		Quiz 11: 10.6
	Thursday	4-Dec	Taylor Series	10.8		
	Friday	5-Dec	Taylor Series	10.8		
	Saturday	6-Dec				
	Sunday	7-Dec				
15	Monday	8-Dec	Polar Coordinates and Complex numebrs	11.3		Achieve 10.8
	Tuesday	9-Dec				Written 13: 10.8
	Wednesday	10-Dec	Polar Coordinates and Complex numebrs	11.3		Quiz 12: 10.8
	Thursday	11-Dec	Polar Coordinates and Complex numebrs	11.3		
	Friday	12-Dec	Polar Coordinates and Complex numebrs	11.3	Classes End	Written 14: Polar & Complex
	Saturday	13-Dec				
	Sunday	14-Dec				
16	Monday	15-Dec	**Week of Finals**		Cummulative Final Exam	
	Tuesday	16-Dec				
	Wednesday	17-Dec				
	Thursday	18-Dec				
	Friday	19-Dec				